

Evidence Before Improvement: A White-Box Contract Audit for Autonomous MLIP Research

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Abstract

Autonomous research agents can optimize a reported metric while silently losing the evidence needed to justify the result. We preregister a deterministic white-box contract-coverage audit of an artifact-bound research trace for machine-learning interatomic potentials (MLIPs). Twelve hand-authored fault operators were each applied to four trace instantiations, yielding 48 single-fault cases and four clean controls. The semantic grader detected 48/48 declared corruptions with no clean rejection; a mutable manifest-only integrity ablation detected 4/48. A separate, previously verified two-iteration MACE replay on an NVIDIA L4 illustrates the policy consequence: one candidate crossed the frozen metric-improvement threshold but remained rejected after its exact-equality reproducibility gate failed. These observations support an infrastructure claim about coverage of the tested contract and fault catalog, not general grader robustness, improved MACE performance, or scientific validity.

1. Introduction

Language-model agents can plan, execute, and write machine-learning experiments (Huang et al., 2024; Yamada et al., 2025), while recent benchmarks evaluate whether agents can reproduce research artifacts (Starace et al., 2025). Success metrics alone, however, do not establish that an experimental conclusion is reproducible: reproducible code and reproducible findings are distinct goals (Bouthillier et al., 2019). This distinction is acute for MLIPs, where model identity, energy and force units, data partitions, hardware, and validation protocols all affect interpretation (Battista et al., 2022; Maxson et al., 2024).

We study a narrower question: *does semantic re-derivation*

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detect repository-defined trace faults more completely than producer-declared hashes? Drawing on fault injection as a controlled dependability-analysis technique (Hsueh et al., 1997), our contribution is a preregistered white-box contract audit over 12 fault families and four seeds, paired with a real-MACE negative-result case study. The paper intentionally stops at an infrastructure claim ceiling.

2. Method and Agent Workflow

Evidence contract. The controller emits sealed objective, proposal, legality, execution, evaluation, decision, lesson, lineage, and manifest artifacts. The evaluator has a distinct code identity, and the decision is recomputed from frozen acceptance constraints. Reflection and reviewer text may influence a later proposal but cannot support a numerical claim. This follows the broader principle of falsification-oriented automated analysis (Huang et al., 2025), but our grader is deterministic and does not use an LLM.

3. Experimental Protocol

Preregistered comparison. For each seed from 20260712–20260715, we generated one clean two-iteration synthetic trace and then made twelve separate derived copies. Each copy received exactly one frozen fault: missing Tea Time, missing lesson, evaluator identity collision, immutable-input change, forbidden label payload, unauthorized remote attestation, decision–metric inconsistency, internal-seal tamper, registered-artifact corruption, inflated scientific status, forged rejection without rollback, or broken lineage parent. The recorded execution evaluated every seed–fault pair once and reports `retry_count=0`. The integrity ablation verifies every path, size, and SHA-256 declared in a mutable producer manifest. For semantic forgeries, that manifest is refreshed intentionally; hashes prove byte identity, not whether the bytes describe a legal experiment. The candidate additionally re-derives cross-artifact constraints.

The primary endpoint is macro fault-detection recall. Success required recall ≥ 0.90 , no rejection of a clean trace, and an absolute recall gain ≥ 0.25 . The benchmark used local CPU only and accessed no scientific labels, protected

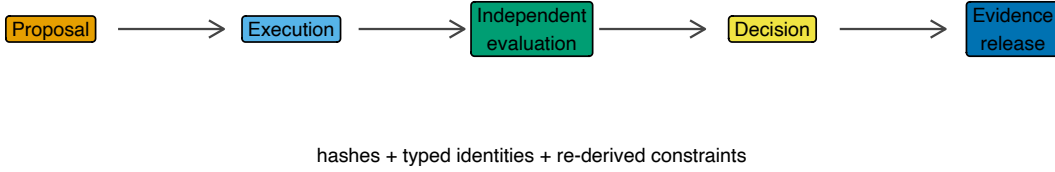


Figure 1. The evaluated workflow binds proposal, execution, independent evaluation, decision, and evidence release. The semantic grader checks both artifact integrity and cross-stage constraints; agent prose and Tea Time reflection are excluded from evidence.

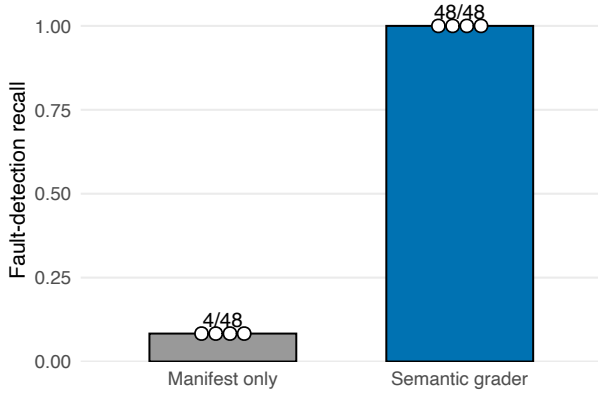


Figure 2. Fault-detection recall over 48 preregistered corrupt traces (four seed-level points per method; 12 families per seed). Bars show aggregate exact counts, and no uncertainty bars are drawn because the finite catalog is reported without population inference. Both methods accepted all four clean controls.

Table 1. Frozen fault-injection result.

Method	Detected	Recall	Clean rejected
Manifest hashes	4/48	0.083	0/4
Semantic grader	48/48	1.000	0/4

partitions, acquisition labels, or remote resources.

4. Results

The semantic grader detected 48/48 corruptions (macro and micro recall 1.00) and accepted 4/4 clean controls. The manifest-only integrity ablation detected 4/48 (macro and micro recall 0.083), for a preregistered macro-recall gain of 0.917 (Table 1); macro and micro values coincide because every family has four cases. Every threshold was met. No clean rejection was observed in four controls, which is not a precise estimate of a population false-positive rate. Because the faults were white-box and repository-defined, the result demonstrates coverage of this contract, not robustness to arbitrary attacks.

Real-MACE negative case. We also reused a hash-verified, policy-attested exact-commit replay on one NVIDIA L4. It used `mace-torch` 0.3.16, a candidate-only checkpoint, a bounded approved label view, one optimizer step per iteration, and an independent aggregate evaluator. For validation force-component MAE in $\text{eV \AA}^{-1}$, iteration 1 improved by 0.053%, below the frozen 0.1% threshold, and was rejected. Iteration 2 improved by 0.133% and passed the metric threshold, but both iterations had `rerun_metric_delta=1.0` and failed the exact-equality reproducibility constraint, so both were rejected. Here 1.0 is a binary mismatch indicator, not the physical magnitude of numerical drift. Exact equality is especially conservative because numerical reproducibility can vary across software releases, platforms, and CPU/GPU execution even with identical seeds (PyTorch Contributors, 2026). The completed trace remained `infrastructure_only` and claim-ineligible.

This case demonstrates the policy consequence: a metric-only rule would have promoted iteration 2, whereas the frozen multi-constraint rule retained the negative result. It does not show that the candidate was scientifically worse, because the current binary rerun endpoint cannot distinguish harmless floating-point variation from decision-relevant instability.

5. Limitations and Conclusion

Our fault operators were written against the grader contract, so perfect recall may overestimate coverage of unanticipated corruption. Four clean controls do not tightly bound false-positive probability. The mutable manifest-only integrity ablation is deliberately narrow, and the experiment evaluates trace validity rather than scientific truth. The MACE replay contains two infrastructure iterations on one device and no frozen-test release, physical-property validation, independent provider replication, or approved checkpoint promotion. Repeated validation access can also create adaptive bias (Dwork et al., 2015); this campaign makes no claim about that risk.

Within these limits, the experiment supports one conclusion: for the twelve preregistered fault families, semantic re-derivation caught failures that producer-declared hashes usually could not, without rejecting the clean controls. In the accompanying MACE case, the same fail-closed posture preserved a nominal metric improvement as a rejection rather than converting it into an unsupported performance claim. Future work should replace the binary rerun flag with tolerance-aware, decision-calibrated deltas and evaluate externally designed faults.

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Technical Appendix: White-Box Contract Audit for Autonomous MLIP Research

Internal evidence edition

12 July 2026

1 Frozen Research Contract

The preregistration froze four seeds, twelve single-fault families, zero retries per case, local CPU execution, and no remote or protected-data access. The primary endpoint was macro fault-detection recall. Success required semantic macro recall at least 0.90, no observed rejection among the four clean controls, and at least 0.25 absolute macro-recall gain over a mutable manifest-only integrity ablation. The canonical specification content hash is `a3361385e420d1692b5b10f87cdb0ffb122a4f08125cfed2ddcc513406819e`.

2 Fault Catalog and Frozen Outcomes

The fault families were missing Tea Time, missing lesson, evaluator identity collision, immutable-input change, forbidden label payload, unauthorized remote attestation, decision-metric inconsistency, internal-seal tamper, registered-artifact corruption, inflated scientific status, forged rejection without rollback, and broken lineage parent. Each family was applied once to a separate derived trace for each seed.

Table 1: Aggregates independently derived from the immutable ledger.

Method	Detected	Corrupt total	Macro recall	Clean rejected
Mutable manifest hashes	4	48	0.0833	0/4
Semantic contract grader	48	48	1.0000	0/4

The ledger hash is `f6bbd45076e63c75bb539fa72439413214bb7234f9199669133032cd3995e2ed`. Independent review found that the initial summary expressed the recall gain with micro quantities and omitted the manifest macro recall, although the preregistration named macro recall. Amendment A-001 recomputed only summary metadata from the frozen ledger. Because all twelve families have four cases, macro and micro recalls coincide and the numerical decision did not change. No case was rerun. The corrected summary hash is `258bdaaedada3dc621b21f62eda754e77be7cd610f3e982c768cfed59384e65b`.

3 Real-MACE Negative Case

The internal source bundle is `artifacts/auto_research/ralphthon-mace-ec2d804/`, generated at exact commit `ec2d804fb3b6dafa012620d83436007569de5e7c`. The anonymous paper calls it `remote-trace-A`. The run was policy-attested on one NVIDIA L4 and used `mace-torch` 0.3.16, a candidate-only checkpoint, an approved bounded label view, and one optimizer step per iteration. Its trace-grade, completion, attestation, dataset, checkpoint, environment, operation, evaluation, and decision artifacts are enumerated with SHA-256 hashes in `private_evidence_map.json` and the public blinded evidence map.

Iteration 1 changed validation force-component MAE by 0.053%, below the frozen 0.1% threshold. Iteration 2 changed it by 0.133% and crossed that threshold. Both were rejected because the exact-equality reproducibility flag failed. The recorded value `rerun.metric.delta=1.0` is a binary mismatch indicator, not an error magnitude. The trace remained `infrastructure_only` and `claim_eligible=false`.

4 Reproduction and Audit

The local fault audit and derived analysis require no network call or secret:

```
env -u PYTHONPATH conda run -n mlip-research-agent \  
  python scripts/research/run_trace_fault_audit.py \  
  --output artifacts/reproductions/free-ralph-trace-audit  
env -u PYTHONPATH conda run -n mlip-research-agent \  
  python scripts/research/build_trace_audit_analysis.py \  
  --benchmark artifacts/reproductions/free-ralph-trace-audit \  
  --output artifacts/reproductions/free-ralph-trace-analysis  
Rscript paper/figures/render_figures.R "$PWD"
```

The submitted ledger is the original execution ledger. Reproduction creates a new ledger for comparison; it must not overwrite the frozen evidence during audit. Decisions and summary endpoints reproduce, while per-row elapsed timings are host-dependent. Amendment A-002 therefore excludes the under-specified `wall_seconds` secondary endpoint from interpretation. The paper was compiled with `pdflatex -no-shell-escape`; all citations and references resolved, all fonts are embedded, and the three-page letter-size PDF was visually inspected page by page.

5 Claim Ceiling

The local fault counts are T1 infrastructure facts, while the reused L4 observations are T2 remote-infrastructure observations. There is no T3–T5 scientific claim. In particular, the evidence does not establish an improved potential, general adversarial robustness, a physically significant rerun delta, or a population false-positive rate of zero.